The examination is being carried out on the **following application documents**

Description, Pages

1-23 as originally filed

Claims, Numbers

1-18 as originally filed

Drawings, Sheets

1/13-13/13 as originally filed

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1 The following document has been indicated in the search report as P-document:

D1 FALCONI FRANCO ET AL: "A novel robust control strategy for industrial systems with time delay", 2021 5TH INTERNATIONAL CONFERENCE ON CONTROL AND FAULT-TOLERANT SYSTEMS (SYSTOL), IEEE, 29 September 2021, pages 372-377, XP034016463, DOI: 10.1109/SYSTOL52990.2021.9595645 [retrieved on 2021-10-29]

It is not to be regarded as state of the art according to Article 54(2) EPC, as the date of priority claimed can be allowed for the relevant parts of the present application (Articles 54(2) and 89 EPC).

2 The application does not meet the requirements of **Article 84 EPC**, because **claims 1, 9 and 18** are not clear.

2.1 Claim 1 is not supported by the description as required by Article 84 EPC, as its scope is broader than justified by the description and drawings.

The present application is directed to a easy-tunable robust controller based on the Smith predictor and a robust RST controller design (see § 30, last sentence or fig. 3 and 4).

However, neither a Smith predictor nor a RST controller are defined in claim 1.

Moreover, since it is undefined how the adjustable single parameter is related to the rest of feature of claim 1, it is unclear how the function of tuning needs to be performed.

In order to overcome these objections, it is suggested to define that the predictive control algorithm is a Smith predictor and to introduce the subject-matter of claim 2 into claim 1.

Claims 9 and 18 should also be amended accordingly.

2.2 In claim 1, it is undefined and unclear how non-linearity or time delay is compensated by using a "response model".

Therefore, the functional statements of claim 1 do not enable the skilled person to determine which technical features are necessary to perform the stated functions.

In order to overcome this objection it is suggested to the applicant to introduce the subject-matter of claim 3 into claim 1.

Claims 9 and 18 should also be amended accordingly.

2.3 According to the Guidelines for Examination in the EPO (see F-III, 8), incorporation of subject-matter by reference to the content of a whole document is not allowed. Thus, the Applicant is requested to remove the incorporation of documents by reference in the description (see § 1), or to expressly incorporate the essential matter from the referred document.

2.4 The vague and imprecise statement in the description on § 90 implies that the subject-matter for which protection is sought may be different to that defined by the claims, thereby resulting in a lack of clarity of the claims (Article 84 EPC) when the description is used to interpret the claims (see Guidelines F-IV, 4.4). This statement should therefore be amended to remove this inconsistency.

3 Because of the reasons stated in section above, independent claims 1, 9 and 18 are not allowable.

The applicant's attention is drawn to the fact that unclear features cannot be used to distinguish an invention from the prior art (T 1969/08; GL F-IV, 4.6).

However, it seems that if these independent claims were clarified as suggested in section above, the subject-matter of these claims would be novel and inventive. It is therefore suggested to draft one new independent claim per category along these lines.

4 Furthermore, upon continuation of the procedure, the applicant is invited to take account of the following points:

4.1 Rule 43(7) EPC, reference signs in the claims;

4.2 Rule 50(1) EPC, documents replacing documents should meet the requirements of Rule 49(5) and (8) EPC; in particular, the EPO will no longer accept handwritten amendments (OJ 12/2013, 603); and

4.3 Rule 42(1)(b) and (c) EPC, conformity of the description to the new claims and reference to the background art. In particular, the applicant should amend the description as needed to fulfil the following requirements:

- The technical field of the invention (Rule 42(1)(a), GL F-II.4.2) must correspond to the amended set of claims.
- The summary of the invention (Rule 42(1)(c) and GL F-II.4.5) must correspond to the amended set of claims. If appropriate, statements such as “The invention is set out in the appended set of claims“ or “The invention is as defined in claim X“ can be applied instead of repeating the claims verbatim.
- All embodiments which are not covered by the claims anymore must be deleted, unless these embodiments can reasonably be considered to be useful for highlighting specific aspects of the amended subject-matter. In such case, they must be prominently marked as not being part of the present invention (T 1808/06):
  - For example, independent claim <X> has been amended by limiting it to the feature of <feature A>. Ways of carrying out the invention using features presented as alternatives to this feature must either be deleted or the fact that they are not encompassed by the claims must be prominently stated.
  - Any combinations of features which do not fall under the scope of the amended independent claims must be prominently marked as not being part of the present invention. For example, amended claim

<X> comprises features < A>, < B> and < C> and the description on page X refers only to B+C as an embodiment of the invention. It must be marked as not being part of the invention.

- Statements such as “The following examples/aspects/embodiments X, Y, Z are not according to the invention and are present for illustration purposes only” may be used at the start of the description (or in the list of figures) when such a situation arises.

- All technical features of the independent claims shall not be presented as optional in the description. Terms such as “for example“, “may“, “can”, “exemplary“, “optionally“, “preferably“ and the like should be removed when preceding a feature of an independent claim.

The division is aware of decision T 1989/18. That decision is, however, not in line with well-established case law of the boards of appeal (T 11/82, T 977/94, T 300/04, T 698/01, T 323/05, T 1808/06, T 1252/11, T 1883/11, T 1646/12, T 237/16, T 1399/17, T 1941/18, T 1494/17) according to which Article 84 EPC requires the adaptation of the description to the claims. T 1989/18 is an isolated decision. For these reasons, the division does not follow T 1989/18. The practice set out in Guidelines F-IV, 4.3 applies.
